

```
[1]: import numpy as np
import math as ma
import pandas as pd

[2]: df = pd.read_csv('/home/subramanyaprasad/Machine Learning/Data Analysis/Notes/03-Pandas/tips.csv')

[3]: df

[3]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251
...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657
240	2718	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672

244 rows x 11 columns

```
[4]: df['total_bill']

[4]:
```

```
0      16.99
1      10.34
2      21.01
3      23.68
4      24.59
...
239    29.03
240    2718
241    22.67
242    17.82
243    18.78
Name: total_bill, Length: 244, dtype: float64

[5]: mycols = ['total_bill', 'tip']
df[mycols]

[5]:
```

	total_bill	tip
0	16.99	1.01
1	10.34	1.66
2	21.01	3.50
3	23.68	3.31
4	24.59	3.61
...	...	...
239	29.03	5.92
240	2718	2.00
241	22.67	2.00
242	17.82	1.75
243	18.78	3.00

244 rows x 2 columns

Instead of this, we can directly give inside 'df' but with double [] - as a list

```
[6]: df[['total_bill', 'tip']]

[6]:
```

	total_bill	tip
0	16.99	1.01
1	10.34	1.66
2	21.01	3.50
3	23.68	3.31
4	24.59	3.61
...	...	...
239	29.03	5.92
240	2718	2.00
241	22.67	2.00
242	17.82	1.75
243	18.78	3.00

244 rows x 2 columns

What is the % of the tip in accordance to the total bill?

```
[7]: (df['tip'] / df['total_bill']) * 100

[7]:
```

```
0      5.944673
1     16.054159
2     16.658734
3     13.978041
4     14.680765
...
239    20.392697
240    7.358352
241    8.822232
242    9.820426
243    15.974441
Length: 244, dtype: float64

Instead, use NumPy

[8]: np.divide(df['tip'], df['total_bill']) * 100

[8]:
```

```
0      5.944673
1     16.054159
2     16.658734
3     13.978041
4     14.680765
...
239    20.392697
240    7.358352
241    8.822232
242    9.820426
243    15.974441
Length: 244, dtype: float64

Now, let's add this series into the DataFrame - new column

[24]: df['tip_percentage'] = np.divide(df['tip'], df['total_bill']) * 100
df

[24]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID	tip_percentage
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959	5.944673
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608	16.054159
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458	16.658734
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260	13.978041
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251	14.680765
...	...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657	20.392697
240	2718	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766	7.358352
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880	8.822232
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17	9.820426
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672	15.974441

244 rows x 12 columns

If we try to create a new column which already exists, it doesn't create new column but it overrides the existing values

Let's reduce the float of tip_percentage

```
[10]: df['tip_percentage'] = np.round(df['tip_percentage'], 2)
df

[10]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID	tip_percentage
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959	5.94
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608	16.05
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458	16.66
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260	13.98
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251	14.68
...	...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657	20.39
240	2718	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766	7.36
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880	8.82
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17	9.82
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672	15.97

244 rows x 12 columns

Let's talk about removing columns/rows -- axis = 1 → columns and axis = 0 → rows

```
[25]: df.drop('tip_percentage', axis = 1)

[25]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251
...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657
240	2718	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672

244 rows x 11 columns

Now we see that tip_percentage is removed - only 11 columns left - IT DOESN'T REMOVE IN THE ORIGINAL DATA

```
[13]: df

[13]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID	tip_percentage
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959	5.94
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608	16.05
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458	16.66
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260	13.98
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251	14.68
...	...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657	20.39
240	2718	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766	7.36
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880	8.82
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17	9.82
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672	15.97

244 rows x 12 columns

To make the changes permanent, we have two options

```
[26]: df = df.drop('tip_percentage', axis = 1) #the simplest
df

[26]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251
...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657
240	2718	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672

244 rows x 11 columns

Now, we can see that tip_percentage is removed from the original csv file

```
[27]: df.shape

[27]: (244, 11)

[ ]:
```